Quiz 1- Section 2

Important Notes: By submitting this exam, you affirm that you have not given or received any unauthorized assistance during this test, and that all work is your own. Academic integrity is highly valued in this course, and I am committed to ensuring a fair testing environment for everyone. Cheating undermines trust and devalues the hard work of others. Be advised that any form of cheating will not be tolerated and will result in disciplinary action, up to and including failure of the course.

Exam Guidelines

- This exam is closed book and notes and **the use of the internet is strictly prohibited**! However, you are free to use a hand-written, double-sided, A4 size "cheat" sheet with your notes.
- **Communication among students is not allowed** during the exam. This includes talking, text messaging, emailing, chatting, forum discussions, etc. Headphones are also not permitted.
- All questions, irrespective of their nature, should be directed only to Prof. Kaya or the teaching assistants.

Submission Guidelines and Deadlines

- When the exam concludes, all papers and the handwritten cheat sheet must be submitted to the TAs or Dr. Kaya.

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A nutritionist at a food research lab is developing a new type of multigrain flour by blending two grains, Grain 1 and Grain 2. Each grain has different nutritional properties and costs, as shown in the table below.

	Grain 1	Grain 2
Starch (%)	30	20
Fiber (%)	40	60
Protein (%)	20	15
Gluten (%)	10	5
Cost (cents/kg)	70	40

The new flour blend must satisfy the following nutritional requirements: (1) The protein content must be at least 18%. (2) The fiber content must be between 40% and 55%. (3) The gluten content must be at most 8%.

(a) [20 points] Formulate a linear programming model to determine the least-cost blend of grains in order to minimize total cost. Clearly define the decision variables, objective function, and all constraints.

(b) [20 points] Graph the feasible region of the problem in the two-dimensional plane. Clearly label all constraints and indicate the feasible region. Solve the problem graphically and determine:

- the optimal fraction of Grain 1 and Grain 2 in the blend,
- the minimum cost per kilogram of flour.

(c) [15 points] At the optimal solution, identify which constraints are binding and which are non-binding. Briefly explain what this means in practical (real-life) terms.

(d) [25 points] By using the complementary slackness theorem show how the optimality of your solution in part (c) can be verified. Compute the dual of this problem and briefly (verbally and mathematically) explain the role of the theorem in the argument.

(e) [20 points] Suppose the nutritionist can choose from a set of grains G and must satisfy a set of nutritional requirements N. Each grain $g \in G$ has cost c_g and contains $a_{n,g}$ percent of nutrient $n \in N$. Each nutrient $n \in N$ has a required minimum level L_n and maximum level U_n in the final mix.

Write the general linear programming formulation for the least-cost blending problem in closed algebraic form using sets, indices, and parameters.

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Solution

(a) Linear Programming Formulation (20 points)

Decision variables

$$x_1 = \text{fraction of Grain 1 in the blend,}$$
$$x_2 = \text{fraction of Grain 2 in the blend.}$$

Objective function

Minimize total cost per kilogram:

$$\min\ Z = 70x_1 + 40x_2$$

Constraints

Blend proportions:

$$x_1 + x_2 = 1$$

Protein requirement (at least 18%):

$$0.20x_1 + 0.15x_2 \geq 0.18$$

Fiber requirement (between 40% and 55%):

$$0.40x_1 + 0.60x_2 \geq 0.40$$

$$0.40x_1 + 0.60x_2 \leq 0.55$$

Gluten requirement (at most 8%):

$$0.10x_1 + 0.05x_2 \leq 0.08$$

Non-negativity:

$$x_1, x_2 \geq 0$$

- Defining the decision variables in a clear way: 5
  - If they do not write down that it is the **fraction** of grains: -1
- Defining the objective function in a clear way: 5
  - If they do not explain it verbally: -1
  - If there is a mistake in the coefficients: -1 (be careful this mistake may carry for the rest of the solution!)
- Constraints: 10
  - If they do not explain the constraint verbally: -0.5 for each constraint
  - If there is a mistake in the coefficients: -0.5 for each constraint (be careful this mistake may carry for the rest of the solution!)

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(b) Graphical Solution (20 points)

Using the constraint $x_1 + x_2 = 1$, substitute $x_2 = 1 - x_1$ into the remaining constraints.

Protein constraint:

$$0.20x_1 + 0.15(1 - x_1) \geq 0.18 \Rightarrow x_1 \geq 0.60$$

Fiber constraints:

$$0.40x_1 + 0.60(1 - x_1) \geq 0.40 \Rightarrow x_1 \leq 1$$

$$0.40x_1 + 0.60(1 - x_1) \leq 0.55 \Rightarrow x_1 \geq 0.25$$

Gluten constraint:

$$0.10x_1 + 0.05(1 - x_1) \leq 0.08 \Rightarrow x_1 \leq 0.60$$

Feasible interval:

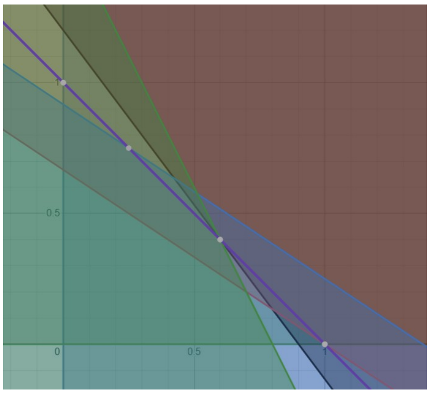
$$0.60 \leq x_1 \leq 0.60$$

Thus, the feasible region collapses to a single point:

$$x_1 = 0.60, \quad x_2 = 0.40$$

Minimum cost:

$$Z = 70(0.60) + 40(0.40) = 42 + 16 = 58 \text{ cents/kg}$$



- Correct graph and labeled constraints: 8
  - If there is a major problem with the graph but they have attempted solving: -3 (be careful this mistake may carry for the rest of the solution!)
  - If they did not label the constraints: -1
- Feasible region identified (single point): 2
- Corner point evaluation: 6
  - If they could not correctly identify the corner point or solve it algebraically : -4
- Optimal solution and cost: 4

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(c) Binding and Non-binding Constraints (15 points)

At the optimal solution $(x_1, x_2) = (0.60, 0.40)$:

Binding constraints:

- Protein constraint
- Gluten constraint
- Blend constraint $x_1 + x_2 = 1$

Non-binding constraints:

- Fiber lower bound
- Fiber upper bound

Interpretation: Protein and gluten requirements are exactly met, meaning they actively restrict the blend. Fiber requirements are satisfied with slack, so they do not influence the optimal mix.

- Binding constraints identified: 6
- Non-binding constraints identified: 4
- Interpretation: 5

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(d) Dual Problem and Complementary Slackness (25 points)

Rewrite the primal problem in standard inequality form (excluding the equality constraint handled via substitution):

Dual variables

$$\begin{aligned} y_P &\geq 0 \quad \text{(protein)}, \\ y_{F_L} &\geq 0 \quad \text{(fiber lower bound)}, \\ y_{F_U} &\geq 0 \quad \text{(fiber upper bound)}, \\ y_G &\geq 0 \quad \text{(gluten)}. \end{aligned}$$

Dual objective

$$\max \; 0.18y_P + 0.40y_{F_L} - 0.55y_{F_U} - 0.08y_G$$

Dual constraints

$$\begin{aligned} 0.20y_P + 0.40y_{F_L} - 0.40y_{F_U} - 0.10y_G &\leq 70, \\ 0.15y_P + 0.60y_{F_L} - 0.60y_{F_U} - 0.05y_G &\leq 40. \end{aligned}$$

Complementary slackness

Since the protein and gluten constraints are binding, their corresponding dual variables may be positive:

$$y_P > 0, \quad y_G > 0$$

Since both fiber constraints are non-binding:

$$y_{F_L} = y_{F_U} = 0$$

Substituting confirms feasibility and optimality, verifying the primal solution via complementary slackness.

Additionally, the dual constraints will become binding as the primal decision variables are strictly positive.

Interpretation: Complementary slackness ensures that only constraints that actively restrict the solution receive positive shadow prices, confirming economic consistency. Strong duality confirms that the optimal value of the dual and the primal are equal to each other.

Alternative Dual Formulation:

$$\begin{aligned} \max \quad & 18y_1 + 55y_2 + 40y_3 + 8y_4 + y_5 \\ \text{s.t.} \quad & 20y_1 + 40y_2 + 40y_3 + 10y_4 + y_5 \leq 70 \\ & 15y_1 + 60y_2 + 60y_3 + 5y_4 + y_5 \leq 40 \\ & y_1 \geq 0 \\ & y_2 \geq 0 \\ & y_3 \leq 0 \\ & y_4 \leq 0 \\ & y_5 \text{ unrestricted} \end{aligned}$$

Note: y_5 is included only for the constraints that had $x_1 + x_2 = 1$ in the dual formulation.

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- Correct dual formulation: 10
  - Minor coefficient or sign mistake: -1
  - Major mistake: - 5
- Identified which variables are strictly positive using complementary slackness: 4
- Identified which variables are zero using complementary slackness: 3
- Identified which dual constraints are binding using complementary slackness: 3
- Clear explanation of complementary slackness to check for optimality: 3
  - Included an explanation, but it's not clear: -1
  - The interpretation is completely wrong: -3

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(e) General Blending Formulation (20 points)

Sets

$$
\begin{aligned}
G &= \text{set of grains,} \\
N &= \text{set of nutrients.}
\end{aligned}
$$

Parameters

$$
\begin{aligned}
c_g &= \text{cost of grain } g \in G, \\
a_{n,g} &= \text{percentage of nutrient } n \text{ in grain } g, \\
L_n &= \text{minimum required level of nutrient } n, \\
U_n &= \text{maximum allowed level of nutrient } n.
\end{aligned}
$$

Decision variables

$$
x_g = \text{fraction of grain } g \text{ in the blend.}
$$

General LP formulation

$$
\min \sum_{g \in G} c_g x_g
$$

subject to

$$
\begin{aligned}
\sum_{g \in G} x_g &= 1, \\
L_n \le \sum_{g \in G} a_{n,g} x_g &\le U_n \quad \forall n \in N, \\
x_g &\ge 0 \quad \forall g \in G.
\end{aligned}
$$

- Sets, parameters, and indices are defined: 6
- Objective function: 6
- Constraints: 6
  - If they forgot the sum $= 1$ constraint: -2
- Clarity and notation: 2